

MAT 426, Class 5 Homework: $PA = LU$ Factorization using GEPP
(Due Tuesday February 24, 2026)

In Exercises 1–3, use Gaussian elimination with partial pivoting to find a $PA = LU$ factorization of A . Clearly identify the matrices L , U , and P

1. $A = \begin{bmatrix} 0 & 0 & 1 & 2 \\ -1 & 1 & 3 & 2 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & -1 & 0 \end{bmatrix}$

2. $A = \begin{bmatrix} 2 & 1 & 3 & -4 \\ 4 & 1 & 0 & -2 \\ 1 & 2 & -3 & 4 \\ 4 & -1 & 1 & 3 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & 2 & 0 & -1 & 1 \\ -2 & -7 & 3 & 8 & -2 \\ 1 & 1 & 3 & 5 & 2 \\ 0 & 3 & -3 & -6 & 0 \\ 3 & -2 & 1 & 0 & 2 \end{bmatrix}$